Electronic Circuit for the Signal Conditioning of PSD Optical Sensors

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***Abstract*— This article presents the methodological process to be followed in the design of signal conditioning circuits for optical sensors (position sensitive detector (PSD) or photodiodes), which generally supply currents of nanoamperes that must be suitably managed before the analog-to-digital conversion process. The first step is to define the requirements for the signal supplied by the sensors and the signal to be sent to the analog-to-digital converter (ADC), taking into account the operating conditions and the configuration of the sensors. The various aspects to be taken into account in the design stages are presented below. In the first stage of the conditioning circuit, the output of the sensors is adapted with a high-gain transimpedance stage and low-pass filtering. In the next stages, the gain is increased to the values required to make maximum use of the dynamic range of the ADC (total gain close to 20M). Two alternatives are studied, one of first order and the other of second order. In both cases, high-pass filtering is performed to eliminate low-frequency optical signal noise, the noise bandwidth is reduced to obtain a flat response in the passband, and the output offset is limited. In the third stage, a gain adjustment is introduced and the appropriate bias is added to the signal to be delivered to the ADC. The design methodology is analyzed using SPICE to verify that the necessary requirements are met. The circuits are then implemented to verify that they behave as required.**

***Index Terms*— Design methodology, instrumentation, photosensors, position sensitive detector (PSD), signal conditioning, visible light positioning (VLP).**

I. Introduction

The problem of indoor location has been a subject of intense study and research in recent years. A number of proposals have been developed to provide solutions to specific applications, with different degrees of accuracy and complexity. It has been accepted for some time that in many indoor activities, knowing the position of the user brings an added value that provides a new set of capabilities for a given specific application.

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The evolution of indoor positioning systems (IPS) can be demonstrated through the analysis of a number of papers which aim to review the current state of the field [1], [2], [3], [4], [5].

A significant research endeavor is currently underway to develop multiple applications, including those that provide support for individuals with special needs, as well as those that track users in large areas, guide people at train stations, airports, museums, and provide security at major events. The development of IPS technologies uses a range of different technologies. The selection of one or more technologies, or the combination of several, depends on the specific requirements of each application. Visible light positioning (VLP) or infrared light-based IPS is currently the least developed. It is therefore recommended that in the development of infrared or visible light-based IPS, which will receive signals from multiple sources with signal strengths of nanoamperes or a few microamperes, a design methodology for the signal conditioning circuits is essential. These circuits must deliver a clean signal at the input of the analog-to-digital converter (ADC), with the maximum span as faithful as possible to the original signal.

In the field of sensor conditioning circuits, numerous papers have been published on the development of such circuits for capacitive sensors. In addition, a considerable number of papers have been published on the development of conditioning circuits for resistive sensors, with a further group of papers focusing on the development of conditioning circuits for capacitive-resistive sensors. Furthermore, a significant number of papers have been published on the development of interfaces for both capacitive and resistive sensors, with the majority of these interfaces being digital. In the case of resistive sensors, work is often focused on bridge-based sensor models.

The work presented in [6] offers a robust signal conditioning circuit for capacitive sensors. The circuit is designed to measure the capacitance of leaky capacitive sensors and uses a phase-sensitive detection technique to mitigate the effect of shunting conductance. Tirupathi and Kar [7] present a design and analysis of a capacitive sensor interfacing circuit. This circuit detects the physical signal as a change in capacitance and provides a voltage output. The developed technique is used to remove low-frequency noise and offset voltage introduced by the operational amplifier. Pal et al. [8] used an OA-based Hartley oscillator circuit to convert a change in capacitance to a change in frequency. The circuit is used for the acquisition of readings from sensors situated at a distance. The output frequency from the circuit is measured and then converted into an equivalent voltage.

Malik et al. [9] present a capacitance-to-voltage converter circuit for measuring the capacitive component of leaky capacitive sensors. A unity gain, frequency-independent quadrature phase shift amplifier is designed for the generation of a phase-shifted signal with high accuracy. The work presented in [10] introduces new and efficient digital signal conditioning techniques for interfacing with resistive sensors. These techniques use a simple circuitry that requires a single reference voltage and single operation cycle and provide a direct-digital output proportional to the sensor resistance. Furthermore, the techniques are enhanced with the addition of a few auxiliary components to address the case of remotely located resistive sensors.

The work presented in [11] proposes an impedance-to-time converter circuit for leaky capacitive sensors that is able to address the aforementioned issues and estimate the capacitor and resistor values. The proposed circuit is a relaxation oscillator, which combines dual slope and charge transfer methods to measure the sensor. In a similar vein, the author proposes in [12] an autonulling-based multimode impedance-to-voltage converter signal conditioning circuit for resistive–capacitive sensors.

Nair et al. [13] propose an efficient digitizer for the measurement of low currents over a wide span. The proposed digitizer technique provides a digital output proportional to the sensed current, with reduced quantization error of the ADC. The design comprises a transimpedance amplifier and a multiregime integrator. In [14], a methodology is introduced to aid in the development of low-cost systems, which avoids measurement inaccuracies. This methodology consists of investigating the errors in the system and further, predicting the systematic errors of each subsystem by modeling its frequency response by SPICE. In [15], an interface circuit for resistive sensors is shown. The circuit offers a number of advantages, including the use of a single-reference voltage in its architecture, the ability to tune or preset the sensor-excitation current, and minimal dependence on many circuit nonideal parameters.

In this context, the methodology used to design signal conditioning circuits is focused on addressing the challenges posed by optical sensors, which exhibit characteristics of dependent current generators and, more specifically, photolateral effect sensors such as position sensitive detectors (PSDs).

This work aims to establish a design process for signal conditioning circuits to be used in IPS based on optical signals. The objective is to enhance the value of systems based on photodiodes, thereby enabling the development of highly precise and cost-effective VLP-based IPS through the utilization of existing lighting infrastructures. The design methodology developed can be extended to numerous other fields, although it is particularly suited to the specific case of PSD signals.

As described by the PSD manufacturer, the position is determined by addition, subtraction, and division operations, with the currents supplied by the anodes. Therefore, it is important to accurately measure these currents, as inaccurate measurements will result in errors in position determination. Consequently, it will not be working with dc signals emitted by LED or IRED bulbs, but with modulated signals. Therefore, the signals provided by the PSD must be carefully managed and processed.

1) Amplify up to several million.
2) Reduce the noise bandwidth.
3) Eliminate any noise from sunlight or low-frequency lighting.
4) Add to the signals the optimal bias to enter the ADCs taking full advantage of the span.
5) Digitize with an appropriate number of bits and quantification noise.

This work will focus on the research and development of the acquisition system and the signal conditioning system for an IPS based on a PSD. The objective of this research is to develop a methodology and process for designing signal conditioning subsystems, so that it is feasible to realize high-precision PSD-based IPS using VLP. For the sake of brevity, stability analysis, impact of component tolerances, effect of component temperature coefficients, influence of suboptimal operational amplifier parameters, electronic noise analysis, and system resolution study will not be described here. Work is underway to investigate these issues.

The design of the requisite circuits necessitates a comprehensive investigation, as they require signals to be subject to substantial amplification while maintaining their original ratio and exhibiting minimal noise. This article presents a detailed analysis of these considerations.

II. Requirements

As a preliminary step in the design process, the specific characteristics of the devices to be used and the final outcome to be achieved must be taken into account. To conditioning the outputs of the PSD sensor in a given working environment, it is necessary to describe the different characteristics of the PSD sensor and the ADC. This will enable the signal to be conditioned properly for conversion, for example, in the ADC of a microcontroller.

A. PSD Sensor

The PSD sensor is a photodiode that has four anodes and a common cathode (Fig. 1). The signal current obtained from these four anodes depends on the impact point of the light emitter on the PSD surface. One of the most important parameters of PSD is the interelectrode resistance (R_{ie}—the resistance between the opposing electrodes of a PSD when it is in a dark state). The interelectrode resistance determines the response speed, position resolution, and saturation photocurrent. The interelectrode resistance is measured with 0.1 V applied across the output terminals of the opposing electrodes while the common electrode is left open. When measuring the interelectrode resistance of a 2-D PSD, the output terminals

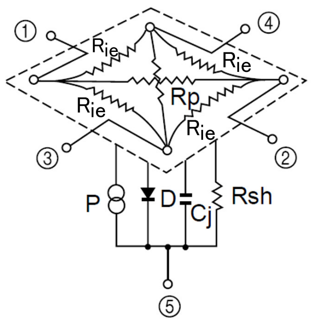

Fig. 1. Equivalent circuitry of a pin-cushion PSD sensor (picture courtesy of Hamamatsu, obtained from the technical information of PSD).

other than the output terminals of the opposing electrodes are also left open. The interelectrode resistance R_{ie} in 1-D PSDs is typically on the order of hundreds of kΩ, while in 2-D PSDs, as in our sensor it is around 10 kΩ.

Position resolution varies with the light level, interelectrode resistance, and size of the PSD (resistance length R_p, usually much larger than R_{ie}). A large interelectrode resistance improves the position resolution; however, the saturation current becomes low, so saturation is prone to occur. The response speed also degrades. The position resolution is defined by "*(noise current/signal photocurrent)* $\times$ *resistance length*." The larger the interelectrode resistance, the smaller the noise current. Therefore, the larger the interelectrode resistance, the better the position resolution. The signal photocurrent is proportional to the incident light level, so the larger the incident light level, the better the position resolution. However, excessive incident light level will cause saturation, so the incident light level must be set within a range where the PSD will not become saturated.

On the other hand, the point of impact can be determined by calculating the photocurrent delivered by the four anodes in accordance with (1) and (2) obtained from [16]. As can be seen, these equations are based on the voltage. A conversion between current and voltage is necessary to obtain the impact point with these equations. (x, y) represents the point of impact of the optical signal on the surface of the PSD, L_x and L_y are the dimensions of the PSD surface, and V_{oi} is the voltage value of each of the anodes after the various stages of conditioning and amplification

$$x = \frac{L_X}{2}\frac{(V_{o2}+V_{o3})-(V_{o1}+V_{o4})}{V_{o1}+V_{o2}+V_{o3}+V_{o4}} \tag{1}$$

$$y = \frac{L_Y}{2}\frac{(V_{o1}+V_{o2})-(V_{o3}+V_{o4})}{V_{o1}+V_{o2}+V_{o3}+V_{o4}}. \tag{2}$$

As can be deduced from (1) and (2), to obtain the position (x, y), it is necessary to use the signals obtained from the currents delivered by the four anodes of the PSD which have been suitably conditioned. It is very important to note that an imbalance in the gain or behavior of the four conditioning channels can result in a significant error in determining the position. This emphasizes the necessity for meticulous consideration in the design of the conditioning and amplification circuitry. Although the selected components are highly precise, these errors, albeit small, will always exist. However, they can be calibrated and corrected once the system is built. For this calibration process, we refer to our previous work [16], where a detailed method for correcting such errors is presented.

In the future, to determine from (x, y) the point in space where the light source is located, it is necessary to have a high degree of precision regarding the parameter f (the focal length of the coupled optical system to the PSD). This parameter will also influence the field of view (FoV) of the system.

Although the focus of this work is not on the IPS but on the conditioning electronics, these equations have been included to assist the reader in understanding the subject matter. As the manufacturer does not provide much information about the response of the sensor, in this work the frequency response has been obtained experimentally. Of these tests, an approximately bandwidth of 200 kHz was obtained (note that these sensors have a very large sensing surface area—9 $\times$ 9 mm^2). The spectral response of the sensor is presented in [17], along with the other PSD parameters. Note that the electrical current provided by the sensor (photosensitivity) is a function of the effective optical signal strength received on its surface and wavelength.

With these parameters, the sensor can be understood sufficiently to approximate the outputs that will be had and to be able to make a proposal of design. As an example, consider an LED emitting warm white light, which has most of its signal strength around 560 nm. The photosensitivity of the PSD for this light will be approximately 0.35 A/W as shown in [17]. Keep in mind that all the signal strength emitted by the emitter is not received by the detector. Only that which falls within the solid angle that the detector covers is received.

B. Analog-to-Digital Converter

The output of the conditioning system to be designed will be connected, for example, to the inputs ADC of the microcontroller. This way, the input signal range of this microcontroller is a requirement to design the conditioning system. The higher resolution will be obtained when the output signal from the conditioning system has the same voltage range as the ADC input. Finally, the requirements due to ADC will be the minimum output voltage V_{o_m} and the maximum output voltage V_{o_M}. In the case studied, a microcontroller was used with an input range of the ADC from 0 to 3 V (consideration should be given to whether unipolar or bipolar signals can be used). However, as will be seen, adaptation to any other range and characteristics of input signals is readily achievable.

C. Working Environment Conditions and Configuration

Once the characteristics of the sensor and the ADC are known, it is time to describe the operating conditions of the system. Considering a Lambertian distribution of the light emitted by LED-based bulbs or spotlights, the signal intensity is considered to be homogeneously distributed on every point of the sensor surface, since the distance between the emitter and the sensor is very large compared with the sensor area (therefore it will not cover a solid angle larger than 0, 01 sr). The irradiance can be considered to be homogeneous over the entire sensor.

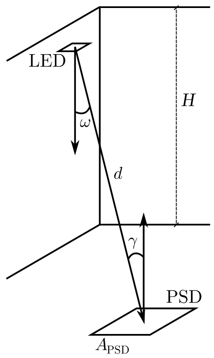

Fig. 2. Diagram showing the relative position between the emitter and the detector.

In general, the signal strength received by the PSD (P_{PSD}) from an LED spotlight can be calculated using the following equation (Fig. 2):

$$P_{\text{PSD}} = I(\omega)\frac{1}{d_{\text{TX}}^2}F(\gamma)R(\gamma)A_x = E(\omega)F(\gamma)R(\gamma)A_{\text{eq}_{\text{PSD}}} \quad (3)$$

where $I(\omega)$ represents the emission pattern, $F(\gamma)$ the transmission function of a filter placed in the detector, d the distance from LED emitter to PSD, $R(\gamma)$ the response of the receiver, and $A_{\text{eq}_{\text{PSD}}}$ the effective area of the receiver (considering the set PSD + lenses). $E(\omega)$ represents the energy per unit area caused by the emitter at the detector's location.

Considering that the emission pattern, $I(\omega)$, follows a Lambertian model with radiation lobe index n (which is the most general case) and the response of the detector + lens assembly follows a model equivalent to a flat lens, P_{PSD} is calculated as follows:

$$P_{\text{PSD}} = \frac{n_{\text{TX}}+1}{2\pi}\frac{1}{d_{\text{TX}}^2}\cos^{n_{\text{TX}}}(\omega)A_{\text{eq}_{\text{PSD}}}\cos(\gamma)P_{\text{LED}} \quad (4)$$

where P_{LED} is the signal strength of the LED emitter.

If the planes of the emitter and the receiver are quasi-coplanar $\omega = \gamma$ and $n = 1$ (the most common case of LED emitters without external lens), P_{PSD} can be obtained from the height between the planes H

$$P_{\text{PSD}} = \frac{1}{\pi}\frac{1}{H^2}\cos^4(\gamma)A_{\text{eq}_{\text{PSD}}}P_{\text{LED}}. \quad (5)$$

One of the main design goals is to avoid saturation of the ADC inputs by trying to use the entire measurement range between V_{o_m} and V_{o_M}. Therefore, it is not the maximum current generated by the PSD that is important, but rather the maximum current that will flow through an anode. The maximum current at an anode occurs when the light beam hits the corner of the PSD. All the current generated by the PSD will flow through the nearby anode. This approximation is valid when working with an emitter under ideal conditions. Cases of multiple emitters should be analyzed using (5) as a function of emitters position. The received signal strength of the PSD can be estimated, e.g., using MATLAB.

III. Signal Conditioning Circuit Design Considerations

Once the system requirements and configuration have been defined, the design of the conditioning system can begin. In this work, there are several aspects to consider.

1) As the PSD output is a current signal, it must first be converted into voltage. As the calculation of the PSD output is approximate and based on ideal parameters, it will be necessary to design according to the principle of not losing information at any stage of the design.
2) The gain will be calculated to cover 95% of the ADC range to avoid the saturation due to unwanted effects.
3) The bandwidth of the conditioning system will depend on the frequencies chosen for the emitters.
4) It will also be necessary to eliminate low-frequency lighting components (e.g., 50 and 100 Hz in Europe and 60 and 120 Hz in America), input offset voltage, and input bias currents.
5) To center the signal with the input range of the ADC, a bias voltage must be added to the output of the conditioning system.
6) The final result of the positioning depends on the relationship between the signal of each PSD channel, and thus the amplification value, in the passband, must be as similar as possible in the four channels. A small variation in the behavior of one channel relative to another can lead to large errors in the result. Therefore, it is necessary to calibrate the behavior and gain balance between different channels.

Note that the phase information is not relevant in this procedure, as we are working with the amplitudes provided by the optical sensors to obtain the AoA. Therefore, in this document we have not made any reference to phase distortions, phase shifts, etc. It should be noted that one conditioning system is required for each PSD channel, although only one is represented here. All the four channels will have the same stages, and an attempt will be made to use the same integrated circuit (IC) with four OAs for the same stage of all the four channels.

Given the low signal strength provided by the PSD (in the order of nanoamperes on many occasions), because the received light power will be very small given the distance to and power of the sources, the amplification required will be several million. It is also important to operate with a single symmetrical supply voltage, given that the signals to be handled are so small. This necessitates the use of highly effective filtering and protection systems to prevent the introduction of noise within the operational band. Consequently, the greater the number of voltages, the more complex and costly the development of the circuit development will be. In addition, an attempt will be made to work with only one type of OA and supply of ±5 V. Keep in mind that working with ±3-V supply signal margin may be lost at the ADC input.

Therefore, the selected operational amplifiers should provide the below characteristics.

1) A very high gain bandwidth product (GBWP).

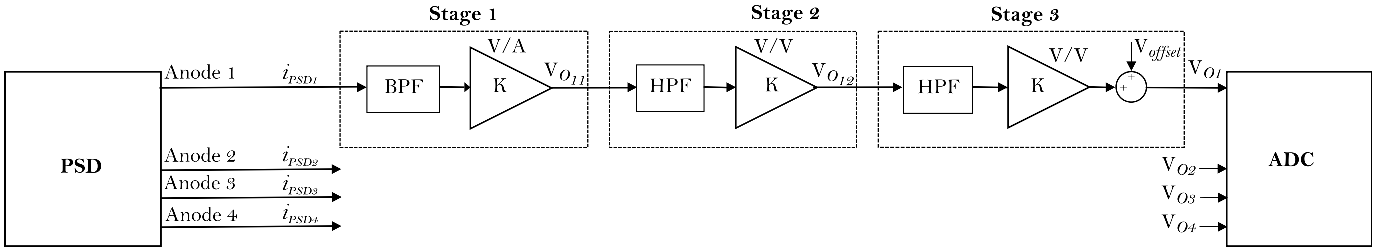

Fig. 3. Diagram of the conditioning system circuit.

2) A very low input noise both current and voltage, due to the small signals handled.
3) It would be desirable to have a low input offset voltage (v_{os}) due to the high gain required.
4) Low input bias currents I_B, ($I_B = ((I_B^+ + I_B^-)/2)$). But it is more important and necessary to have a low input offset current ($I_{\text{OS}} = |I_B^+ - I_B^-|$).
5) High input resistance in differential-mode (R_{id}) and low input capacitance in differential-mode (C_{id}).
6) High open-loop gain (G_o).

It is evident that the preceding paragraph implies a demand for the OA to be ideal in all respects. It is unavoidable that some of these objectives will have to be compromised to achieve others. Nevertheless, in the process of describing the design techniques, it will address the drawbacks that arise when any of the above wishes are not met.

It could be assumed that the use of a single-supply OA would streamline the circuit; however, since the signal is bipolar, the incorporation of a dc level shifter or virtual ground would be required, potentially resulting in a design cost comparable to the use of a dc–dc converter chip that generates a dual supply.

In consideration of the requisite conditions, a three-stage conditioning system has been proposed as illustrated in Fig. 3. This scenario can be studied as a general case when working with this type of sensor.

The first stage consists of a transimpedance amplifier that converts the weak PSD current signal into a voltage. In addition, the signal is amplified by a significant factor. The dc noise from the sensor's dark current and from sunlight is eliminated, and low-frequency signals from artificial lighting are attenuated.

A second stage will amplify the signal to bring it as close as possible to the signal to be provided to the ADC. It should be noted that the input signal in the second stage may also contain low-frequency components (from lighting, etc. that are not fully filtered in the first stage and also offset output noise from the first stage). These components should be amplified once more in the second stage, and the amplifier may become saturated, resulting in the loss of information. It is therefore advisable to introduce a high-pass filter between the first and second stages.

To reach the calculated theoretical amplification and to make final adjustments based on empirical evidence, it may be necessary to include a third stage in our procedure.

It is conceivable that the above adjustments could be made to the second stage, but as this is a high amplification stage and must also include accurate filtering, any small change to one component will require a total redesign of the stage. Consequently, this approach is not deemed appropriate. However, an additional amplification gain in the third stage will not have considerable effects. Furthermore, this third stage will be used to center the signal in the input range of the ADC by means of appropriate bias signal.

Note that between the second and third stages, a high-pass filter must also be introduced to remove the offset from the second stage and low-frequency signal remnants.

Each of the stages shown in Fig. 3 will be described in subsequent paragraphs, indicating the objectives pursued by each stage along with a detailed account of its fundamental characteristics.

A. First Stage

The first stage consists of a transconductance amplifier. To define the circuit design, it is necessary to consider that in addition to the information signal (ac), there will be dc components corresponding to the PSD dark current and the current generated by sunlight or lighting. It should also be noted that the real OA presents offset input voltage (v_{os}) and input bias currents(I_B^+, I_B^-). Fig. 4 illustrates the connection of the PSD equivalent circuit to the first stage of the four channels.

The one-channel circuit, for the purposes of study, is depicted in Fig. 5. The presence of C_{11} serves to prevent the amplification of the input offset voltage of the OA by interrupting the feedback loop and also to prevent the input dc component from reaching the first stage of amplification. Resistor R_{11} is required to properly bias the photodiode (PSD), due to the presence of C_{11} does not allow polarization through the OA. The value of R_{11} has chosen to have a low value to ensure that variations in the current caused by ambient light or sunlight (practically a dc signal) do not significantly alter the polarization point of the PSD, but could be further reduced if necessary.

The expression of the transfer function (TF) of this first stage is shown in (6). In TF analysis, the effect of R_{ie} has been neglected in comparison to the much lower value of R_{11}, which is 298 Ω. Both R_{11} and C_{11} constitute a high-pass filter [see (7)]. The gain is adjusted by the resistor R_{12} which, in conjunction with C_{12}, provides a low-pass filter. The value of C_{12} is selected taking into account the desired bandwidth and system stability. It allows the filtering of high-frequency noise, out of the frequency range to be used in the application. The expressions of the low-pass filter cutoff frequency and

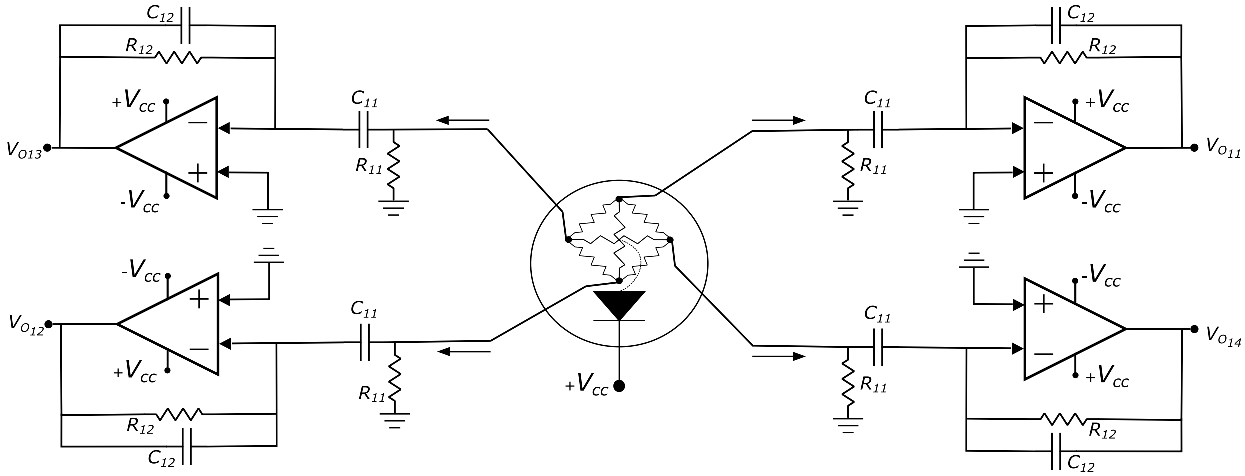

Fig. 4. Connection of the PSD equivalent circuit to the first stage of the four channels.

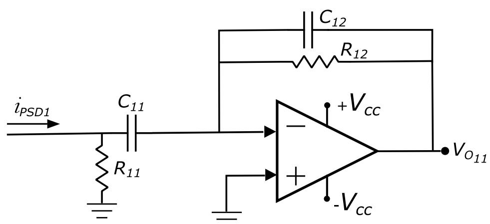

Fig. 5. Circuit of stage 1.

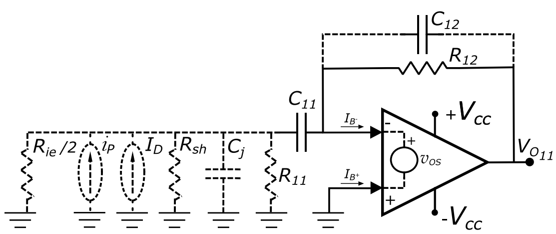

Fig. 6. Circuit of stage 1 to analyze real effects.

gain are shown, respectively, in (8) and (9)

$$\mathrm{TF}_1 = \frac{-C_{11}R_{11}R_{12}s}{(C_{12}R_{12}s+1)(C_{11}R_{11}s+1)} \tag{6}$$

$$f_{cH1} = \frac{1}{2\pi R_{11}C_{11}} \tag{7}$$

$$f_{cL1} = \frac{1}{2\pi R_{12}C_{12}} \tag{8}$$

$$G_1(V/A) = -R_{12}. \tag{9}$$

Analyzing the effects of v_{os} and the I_{B+} and I_{B-} bias currents of the OA in the first stage (Fig. 6), it is found that its effect on the output, $V_{O_{11}\mathrm{offset}}$, is as shown in the following equation:

$$V_{O_{11}\mathrm{offset}} = v_{\mathrm{os}} + R_{12}I_B^- \tag{10}$$

$$V_{O_{11}\mathrm{offset}}|_{T^a} = v_{\mathrm{os}}|_{T^a} + R_{12}I_{B^-}|_{T^a}. \tag{11}$$

It should be noted that the effects on the input offset of CMRR, in (12), and PSRR, in (13), have been neglected because of their low values. The value of V_{CM} is approximately equal to the value of the voltage at the noninverting terminal of the OA and V_{refI} is the value of the voltage of the internal reference of the OA (and CMRR presents very high value). The PSRR effect is usually neglected given that the V_{dc} power supplies of the OA are stable and have no fluctuations. It is recommended that a pair of ceramic capacitors with a capacitance of approximately 0.01 μF be introduced at the power supply terminals of the OAs, with one capacitor on each terminal. In addition, a capacitor with a capacitance of approximately 10 μF should be introduced in parallel, with the purpose of filtering power supply fluctuations

$$\Delta v_{\mathrm{os}}|_{\mathrm{CMRR}} = (V_{\mathrm{CM}} - V_{\mathrm{refI}})/\mathrm{CMRR} \tag{12}$$

$$\Delta v_{\mathrm{os}}|_{\mathrm{PSR}} = \Delta V_{cc}/\mathrm{PSR}. \tag{13}$$

The addition of a resistor at the noninverting input of the OA could help correct the offset due to bias currents. However, it has been decided not to include this as current OA technology allows low input bias currents (furthermore a high-pass filter will also be placed between the first and second stages to prevent the offset output from being passed and amplified). This will greatly simplify the circuit for future noise and resolution analysis.

The values of the parameters in question are highly dependent on temperature, and therefore, if the manufacturer provides us with values for various temperatures these can be used. Otherwise, the values can be calculated from the value for $T^a = 25$ °C by considering and taking into account the average drift values provided by manufacturers in data sheets.

B. Second Stage

The second stage will be a medium-high gain and a high-pass filter.

At this point, it is important to ascertain the bandwidth to be used and the frequency ranges to be used in the proposed application. It is also important to bear in mind that the design of the IPS is based on optical signals and that the lighting in the buildings will typically have frequencies between 50 and 120 Hz. Furthermore, the following factors must be taken into account.

1) Operational amplifiers are limited in GBWP. This may limit the maximum gain of the stage, or it may force lower frequencies to be used in the application.

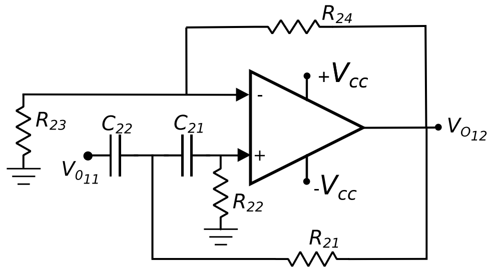

Fig. 7. Circuit of stage 2 using second-order high-pass filter.

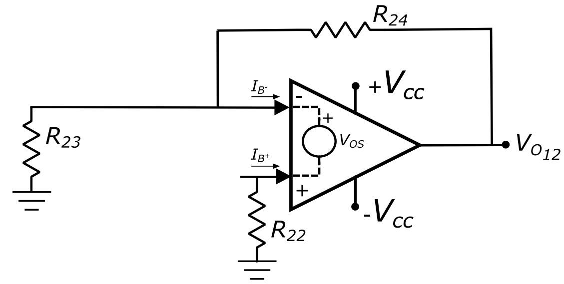

Fig. 8. Circuit of stage 2 for offset dc analysis.

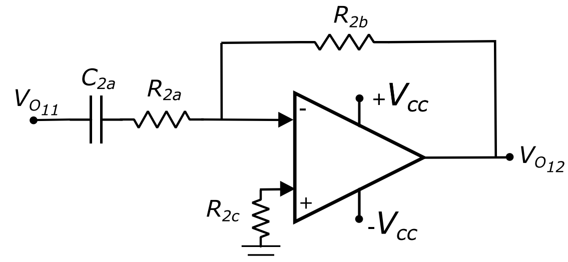

Fig. 9. Circuit for the second stage in the case of using first-order filtering.

2) Frequencies to be used. It is directly related to the above explanation, and sometimes means that they have to be lower.
3) Low-frequency noise that is going to appear in the application.

In the event that the frequencies to be used are significantly higher than the expected noise, a first-order filter with an appropriate cutoff frequency and gain can be implemented far away from these noise frequencies, as it will attenuate them sufficiently. Conversely, in the event that the frequencies to be used are not sufficiently separated from the noise frequencies, a second-order filter must be implemented. The performance of the amplifier will remain constant at the selected frequencies and it will also sufficiently attenuate offset and low-frequency noise, preventing saturation of subsequent stages in the system.

This section presents the development of a first-order inverting stage and a second-order stage of the Sallen–Key topology. Fig. 7 illustrates the second-order solution, and its TF is given by the following equation:

$$\mathrm{TF}_{2a} = \frac{V_{O_{12}}}{V_{\mathrm{in}}}(s) = \frac{\left(1+\frac{R_{24}}{R_{23}}\right)s^2}{s^2 + s\left(\frac{1}{C_{22}R_{22}} + \frac{1}{C_{21}R_{22}} - \frac{R_{24}}{C_{22}R_{21}R_{23}}\right) + \cdots + \frac{1}{C_{22}C_{21}R_{21}R_{22}}}. \tag{14}$$

The expressions for the analysis of the HPF cutoff frequency and gain are presented in the following equations:

$$f_{cH2a} = \frac{1}{2\pi\sqrt{C_{21}C_{22}R_{21}R_{22}}} \tag{15}$$

$$G_{2a} = 1 + \frac{R_{24}}{R_{23}}. \tag{16}$$

In this stage, the value of the output offset voltage assumes considerable significance, given that the signal has already been highly amplified and the addition of offset effects could potentially result in saturation.

In the case of using a stage with second-order filtering, the circuit for offset dc analysis is depicted in Fig. 8.

Note that the dc voltage of stage 1 has been filtered by means of capacitor C_{22}. The dc voltage at the output due to the effects of the v_{os} and the I_B bias currents can be expressed as follows, where A is (R_{24}/R_{23}) and $R_{\mathrm{eq}} = R_{24}||R_{23}$:

$$V_{O_{12a}\mathrm{offset}} = (1+A)\left(v_{\mathrm{os}} + R_{\mathrm{eq}}I_B^- - R_{22}I_B^+\right). \tag{17}$$

From (17), several important conclusions can be obtained to mitigate the effects of offset noise. The first consideration is the importance of selecting OAs with the lowest possible values of v_{os} and I_B which also fulfil the other requirements.

It can also be concluded that if R_{22} is equal to the parallel of R_{23} and R_{24}, the value of the offset output can be limited if the OA presents a saturation current I_{os} small value. In this case, (17) can be written as follows:

$$V_{O_{12a}\mathrm{offset_{max}}} = (1+A)\left(v_{\mathrm{os}} + R_{\mathrm{eq}}I_{\mathrm{OS}}\right). \tag{18}$$

On this occasion, to see the influence on $V_{\mathrm{O_{12}offset}}$ of the CMRR, it must be taken into account that [see (12)] $V_{\mathrm{CM}} \approx V^+ = R_{22}I_{B^+}$ which assumes a second-order error that is negligible. The resistance in the Sallen–Key designs is typically of a low value, and for the same reasons as in stage 1, it is therefore necessary to disregard this effect.

In case of using a first-order approach, the circuit is presented in Fig. 9. Its TF, HPF cutoff frequency, and gain are shown in the following equations:

$$\mathrm{TF}_{2b} = \frac{-C_{2a}R_{2b}s}{C_{2a}R_{2a}s+1} \tag{19}$$

$$f_{cH2b} = \frac{1}{2\pi R_{2a}C_{2a}} \tag{20}$$

$$G_{2b} = \frac{-R_{2b}}{R_{2a}}. \tag{21}$$

Fig. 10 shows the circuit to be analyzed to determine the effect of the offset bias current [see (22)]. It should be noted that the dc input offset voltage is not amplified due to the fact that C_{2a} breaks the amplification branch for the dc signal. In addition, the impact of the bias currents can be significantly reduced by selecting appropriate resistors ($R_{2c} = R_{2b}$). For analogous reasons to those previously discussed in the context of stage 1, the effects on the input voltages of the CMRR and PSRR are going to be neglected

$$V_{O_{12b}\mathrm{offset}} = v_{\mathrm{os}} + R_{2b}I_{B^-} - R_{2c}I_{B^+}. \tag{22}$$

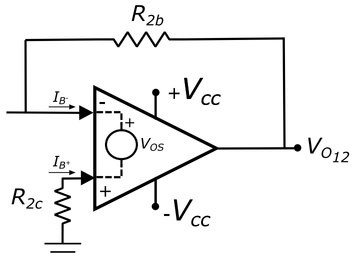

Fig. 10. Circuit of stage 2, first order for offset dc analysis.

C. Third Stage

Finally, a third stage is implemented for adjustments as illustrated in Fig. 11. An inverting amplifier is used to add a bias voltage to condition the signal to the ADC input. In this case, the gain amplification is low to medium allowing the necessary adjustments to be made to the design by simply adjusting the value of the resistor R_{31}.

Note that in this case, a first-order high-pass filter is also implemented, which has the purpose of eliminating the offset noise of the previous stage. It is assumed here that any low frequencies present at the input have already been sufficiently attenuated in the second stage, and therefore a second-order filter is not necessary.

The reference voltage placed on the noninverting input terminal of the OA is used to center the signal in the input voltage range of the ADC, and in this design its value is 1.5 V. The output resistance of this reference must be much lower than R_{33} so that it does not affect the compensation of the polarization currents (I_B). It is important to note that this dc offset cannot be made from the inverting input terminal because it prevents the signals of the four PSD channels from communicating via the reference voltage. To implement the voltage reference, it is suggested to use a 1.5-V fixed voltage reference, such as the ISL21080-1.5V IC. In this case, the max output resistance varies between 0, 1 and 0, 35 Ω.

When choosing the value of the resistors for a given gain, the values should be as small as possible, without the currents being too large, to avoid the effects of the offset being felt at the output and also the currents demanded to the power supply system being large (a balance must be chosen). The following equation shows the TF of stage 3

$$\mathrm{FT}_3 = \frac{-C_{31}R_{31}s}{C_{31}R_{32}s+1}. \tag{23}$$

Values in units of at most a few KΩs at the most are suggested for this stage. The following equations show the expressions for the gain and cutoff frequency of the stage:

$$G_3 = \frac{-R_{31}}{R_{32}} \tag{24}$$

$$f_{cH3} = \frac{1}{2\pi R_{32}C_{31}}. \tag{25}$$

On the other hand, the output offset in stage 3 is expressed in (26). The dc input offset voltage is not amplified by C_{31}. Furthermore, the effects of the bias currents I_{B+} and I_{B-} can be reduced by appropriate resistor choices, as discussed in the previous stages (if $R_{31} = R_{33}$, $V_{O1\mathrm{offset}} = v_{\mathrm{os}} + R_{31}I_{\mathrm{os}}$). For the same reasons as in stage 1 study, the effects on CMRR and PSRR input voltages are neglected

$$V_{O1\mathrm{offset}} = v_{\mathrm{os}} + R_{31}I_{B^-} - R_{33}I_{B^+}. \tag{26}$$

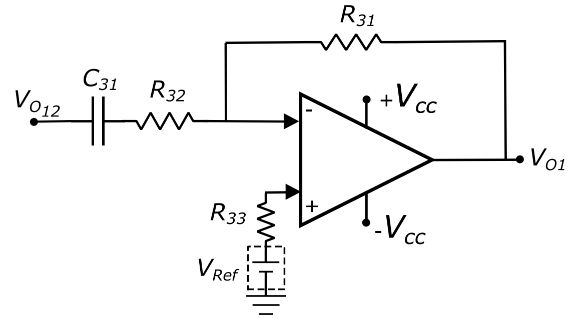

Fig. 11. Circuit of stage 3.

TABLE I

SIGNAL STRENGTH OF THE EMITTER, HEIGHT BETWEEN EMITTER AND RECEIVER, EFFECTIVE AREA, AND FOV OF THE RECEIVER USED IN THE SIMULATIONS

$\mathbf{P_{LED}}$	**H**	$A_{eq_{\mathrm{PSD}}}$	**FoV**
90 mW	2.5 m	0.125e-3 m^2	60° (±30°)

IV. DESIGN AND SIMULATION OF AN APPLICATION

A. Setup to the Design Conditions

The aim of the proposed application is to adapt the analog output signal of a Hamamatsu PSD S5991-01 [17] to the input of the ADC of the LPC546xx microcontroller. The microcontroller in question has an ADC with multiplexed inputs. The ADC operates with 12-bit successive approximation conversion, with a conversion rate of up to 5.0 MHz at 12-bit resolution. The measurement range is between 0 and 3 V, which precludes the use of bipolar input signals. This is an important consideration when designing conditioning systems.

To calculate the total gain of the circuit, it is necessary to determine the maximum current that each anode of the PSD will provide in any situation. This is because the signal must not saturate the ADC input. In this application, it is assumed that the PSD has an effective area of ±4 mm (in reality, it has ±4.5 mm), and that it has coupled a lens of half-inch diameter (equivalent reception area $A_{\mathrm{eq_{PSD}}}$) and 8-mm focal length. Furthermore, it is assumed that the lens has a transmittance of close to 100% and does not reflect energy in the wavelengths under consideration. It is also assumed that the LED light source to be used emits 90 mW of optical signal strength at the peak of the modulation signal, with a Lambertian emitter pattern and with warm white light centered at about 590 nm. Furthermore, it is assumed that the emitter and receiver are in coplanar planes and at a distance of 2.5 m. In accordance with these conditions, the data required for the calculations are presented in Table I.

Assuming that in the FoV there will be only one emission source, the maximum effective signal strength will be received when it is in the vertical of the source, being the effective signal strength, according to (5), of 0.575 muW, generating a current according to the spectral response of the sensor [17] (assuming 0.37 A/W for a wavelength of 590 nm) of 213 nA.

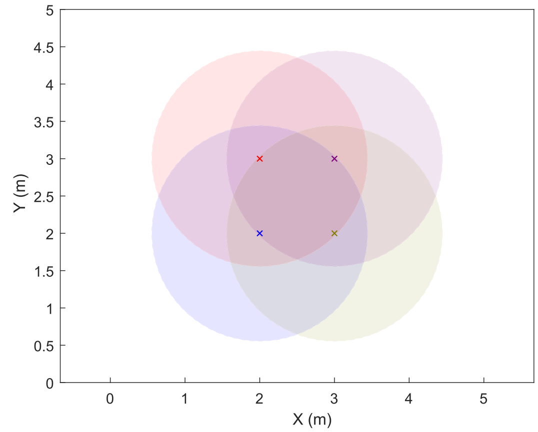

Fig. 12. LEDs disposition layout and their coverage.

It should be noted that the current that would be generated would be divided into four equal parts, with each anode collecting 53 nA. However, this is not the worst case. The most extreme case would be generated when the impact focused by the lens would occur in one of the corners of the sensor, since all the current would be routed to the anode attached to that corner. In this case, the effective signal strength received (with approx. 30° angle) would be 0.32 muW and the current generated would be 120 nA, which would be entirely collected at the anode.

In the present application, however, the objective is to detect multiple signal sources simultaneously (up to four, each emitting a signal with a different frequency). Consequently, the total current to be considered for each anode is the sum of the currents generated by the signal from all the sources. In this case, it is not straightforward to determine which is the worst case mathematically, and therefore a MATLAB simulation has been used. For the purposes of this experiment, it is assumed that the sources are separated by 1 m, forming a square, and the plane in which they are located is separated from the detector by a height of 2.5 m (a schematic is shown in Fig. 12). Taking into account this height and the viewing angle of the detector (30°), the coverage map shown in Fig. 12 has been obtained.

This simulation in MATLAB performs a sweep of the detector position of ±2.5 m in X and ±2.5 m in Y from the center of the four emitters, calculating at each point the current generated at an anode by all the sources. Launching the simulation for the proposed layout, we obtain that there are four peaks near the center of the trace. The value of these peaks is a factor of 1.21 with respect to the maximum value obtained with a single source. Thus, assuming that all the LEDs emit with the same optical signal strength (90 mW), the worst case current will be approximately 120 nA $\times$ 1.21 = 145 nA. Note that this is just a starting point for calculating the system components.

This way, the amplification will be calculated to cover 95% of the ADC range. This has been calculated by obtaining a total gain of 20 000 000 V/A approximately. Regarding the bandwidth, the working frequencies used are between 5 and 14 kHz. It is proposed that a 100-kHz bandwidth be reserved from that frequency onward for future expansions, to avoid high-frequency noises and to allow for the addition of emitters with more frequencies. It is also necessary to eliminate the dc voltage and offset noise in the intermediate stages to avoid system saturation. Moreover, a 1.5-V bias offset should be added at the end of the conditioning system to have an unipolar signal centered in the ADC to avoid overcoming 3 V at the ADC input.

B. Design and Simulations

In this case, the OA proposed is OPA4820 from Texas Instruments [18]. In choosing the OA, the indications given in Section III have been taken into account. Some of the characteristics of this OA are as follows.

1) 4 OA within each chip.
2) Max. power supply $\rightarrow$ ±6.5 V.
3) Small-signal bandwidth $\rightarrow$ 650 MHz.
4) GBWP for high gains $\rightarrow$ 220 MHz $|_{G=+2}$.
5) Input noise $\rightarrow$ 2.5 nV/$\sqrt{\text{Hz}}$ and 1.7 pA/$\sqrt{\text{Hz}}$.
6) Input offset voltage $\rightarrow$ ±0.25 mV.
7) Input bias currents $\rightarrow$ 9 μA.
8) Input offset current $\rightarrow$ 100 nA.
9) Open-loop voltage gain $\rightarrow$ 66 dB.
10) Input impedance differential-mode $\rightarrow$ 18 kΩ, 0.8 pF.
11) Input impedance common-mode $\rightarrow$ 6 MΩ, 1 pF.

Analyzing these characteristics, it is very convenient to be able to work with a ±5 V-power supply and to be able to generate signals that cover the whole dynamic range of the ADC input. This would be challenging to achieve with an OA powered by ±3 V. It is also beneficial to have four OAs in the same package so that all of them (each one in each channel) are affected by the same situations, such as noise. The frequency behavior and input noise are excellent. The GBWP is sufficient for this application. The input offset voltage is not optimal but tolerable as it will be filtered at each stage avoiding amplification. Furthermore, the input capacitance is very small in all the cases.

The presence of elevated bias currents and input offset voltage (more notables in the SPICE models that in data sheet) represents a potential concern. Careful design considerations are required to ensure that input bias currents do not result in high offset voltages at the output. Undoubtedly, the open-loop voltage gain and the differential-mode input resistance of this OA represent the most critical parameters. These two parameters have the potential to significantly decrease the gain of the amplifier, especially if they are not meticulously analyzed and managed throughout the design process. Furthermore, some values of parameters obtained from the manufacturer SPICE models can also be provided: I_{OS} reaches 7.2 μA, G_o 7600, and R_{id} 5K 6 Ω.

This section presents fundamentals and considerations to the design methodology. The use of SPICE simulations will facilitate a deeper understanding of these fundamentals. The calculation of component values and the simulations of the proposed circuit will be presented. The PSD sensor used is modeled as a current source in parallel with a 1-nF capacitor and a 1-GΩ resistor according to the manufacturer's data sheet [17].

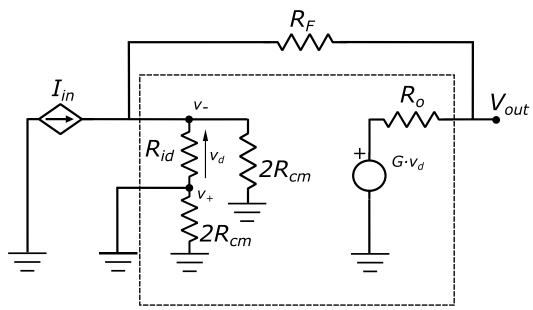

Fig. 13. Equivalent circuit for transimpedance topology analysis.

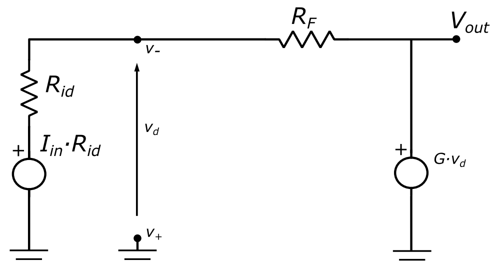

Fig. 14. Simplified circuit of Fig. 13.

The frequency response and output signal are initially analyzed in stages, and then with the complete circuit, in terms of gain, offset, voltage range, and bandwidth. This is done to validate the model and design process.

1) First Stage: The initial stage is a transimpedance amplifier. First, given the low values of G_o and R_{id}, its influence on the gain in the working band will be determined by means of the equivalent circuit, using the real operational amplifier model [19]. Neither the offset voltage nor the polarization inrush currents (whose effects we have already studied) nor the capacitances have been included. Fig. 13 depicts the equivalent circuit for a transimpedance topology with no resistor on the noninverting input terminal. R_{id} represents the input resistance in differential-mode, R_{cm} the input resistance in common-mode, R_F the feedback resistance, R_o the OA output resistance, and G_o the open-loop gain of the OA.

Fig. 14 presents the circuit, after considering the following assumptions: R_{cm} is more than three orders of magnitude larger than the other resistors, R_o is negligible compared with R_F, and $G_o \gg 1$. From the circuit in Fig. 14, the gain in this transimpedance topology is not reduced, as shown by the following equation:

$$\frac{V_{\text{out}}}{I_{\text{in}}} = \frac{R_{\text{id}} R_F}{(1 - 1/G_o)(R_{\text{id}} + R_F) - R_F} \approx R_F. \tag{27}$$

Once this has been verified, the model for SPICE provided by the manufacturer will be used to simulate the frequency behavior of the device under consideration at different gain conditions, without the application of filtering. This test will enable the determination of the effects of the real frequency behavior. Fig. 15 shows the behavior for 50, 10k, 30k, and 50k gains. It can be observed that an instability emerges at frequencies of 30, 2.5, 1.5, and 1.125 MHz, respectively. Consequently, in addition to filtering noise from the unused bandwidth, as proposed in Section III, a low-pass filter must be implemented in this initial stage. The values of the components

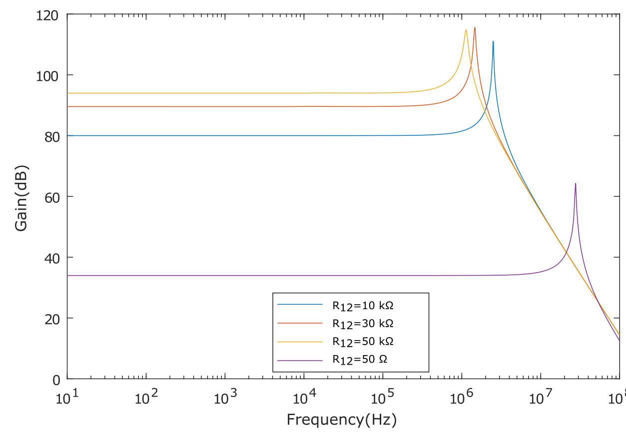

Fig. 15. First stage response (transimpedance amplifier) according to different R_{12} values.

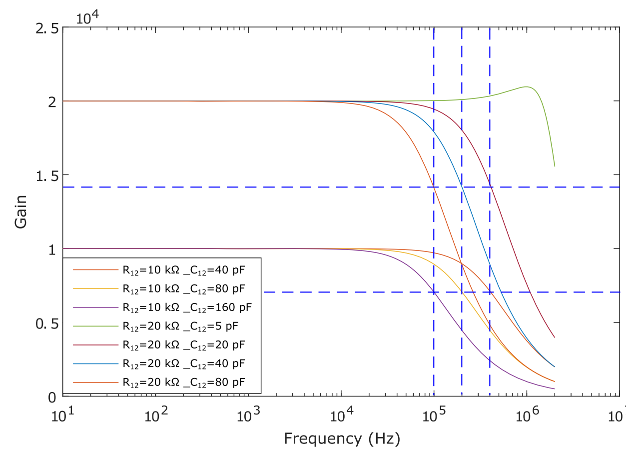

Fig. 16. First-stage response (transimpedance amplifier) according to different R_{12} and C_{12} values.

in the circuit of the first stage (Fig. 5) have been calculated using (7)–(9).

Fig. 16 shows the stage simulation including an LPF with cutoff frequency of 100, 200, and 400 kHz and 10 and 20k gains (to vary the cutoff frequency, the filter capacitance value has been modified). Note that the bandwidth increases with decreasing capacitance, as expected. This can be used for a system requiring higher bandwidth. However, the capacity cannot be decreased as much as desired, because instabilities will occur. As an example, it is shown that with a gain of 20k when the capacitance is reduced to 5 pF instabilities already appear.

It is essential to ensure that the operating frequencies are within the flat region of the filter. In the event that this is not the case, it is necessary to either reduce the filter's gain or increase its bandwidth. Finally, for the circuit in Fig. 5, a 20-kΩ resistor and a 19=pF capacitor have been selected for further implementation and testing purposes (selected from standardized values of commercial components). For R_{11} and C_{11} components, values of 298 Ω and 6.8 μF have been proposed. Fig. 17 shows the frequency response of the entire first stage, with the components as indicated.

2) Second Stage: Before starting with the design of this second stage, we will analyze how the fact that the OA presents low values of G_o and R_{id} affects a voltage amplifier stage.

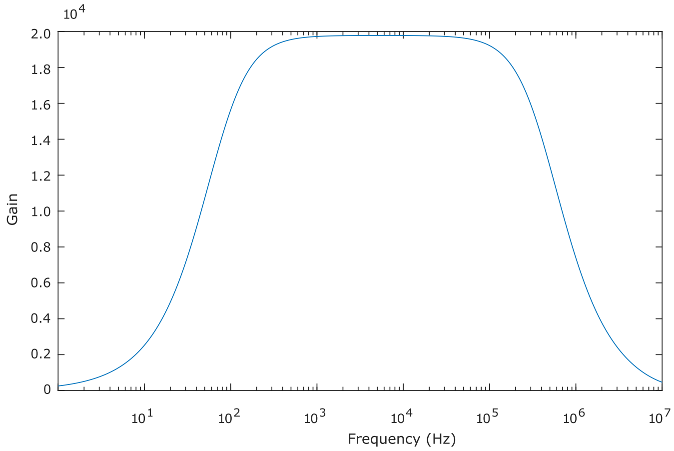

Fig. 17. Frequency response of the entire first stage.

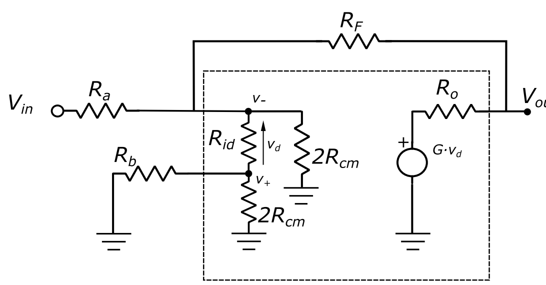

Fig. 18. Equivalent circuit to analyze the gain in a voltage amplifier inverting topology.

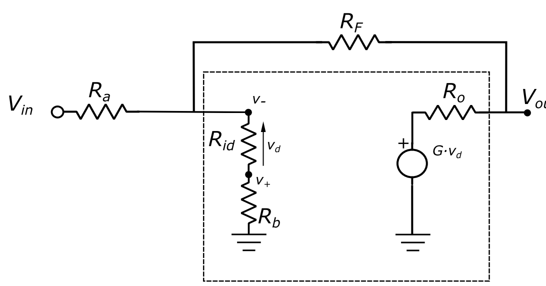

Fig. 19. Simplified circuit of Fig. 18.

An inverting amplifier configuration as shown in Fig. 9 has been considered. Fig. 18 shows the equivalent circuit, using the real OA modeling as was done in the transnimpedance stage. Assuming that R_{cm} is much larger than the rest of the resistors, which is true, and that R_F will be much larger than R_o, the circuit could be drawn as shown in Fig. 19.

Starting from the circuit of Fig. 19, we obtain that the gain of this stage is

$$\frac{V_{\mathrm{out}}}{V_{\mathrm{in}}} = \frac{G_o R_{\mathrm{id}} R_F}{(R_{\mathrm{id}} + R_b)(R_a + R_F) + R_a R_F + G_o R_{\mathrm{id}} R_a}. \tag{28}$$

In the case that R_{id} tends to infinity, which would be the behavior of an ideal OA, we could write

$$\frac{V_{\mathrm{out}}}{V_{\mathrm{in}}} = \frac{G_o R_F}{(R_a + R_F) + G_o R_a} \approx \frac{R_F}{R_a}. \tag{29}$$

To analyse the situation, we will assume that G_o is not very large and R_{id} is of medium value. If we additionally equalize the value of the resistor $R_b = R_F$ (to reduce the effects of the input bias currents to the output offset, assuming there is a capacitor between the first and second stages), the remaining expression is

$$\frac{V_{\mathrm{out}}}{V_{\mathrm{in}}} = \frac{G_o R_{\mathrm{id}} R_F}{(R_{\mathrm{id}} + R_F)(R_a + R_F) + R_a R_F + G_o R_{\mathrm{id}} R_a}. \tag{30}$$

As can be seen, the gain reduction can be very significant. For example, if $R_a = 100\ \Omega$ and $R_F = 75\ \mathrm{k}\Omega$ resistors were used with this amplifier, the gain would be reduced by more than half. To be close to the ideal value (R_F/R_a), the following should be met

$$\frac{(R_{\mathrm{id}} + R_F)(R_a + R_F)}{R_a} + R_F \ll G_o R_{\mathrm{id}}. \tag{31}$$

Thus, for example, with the OA being used, if the value of R_a were chosen to be 100 Ω, the rest of the resistors in the circuit should not be greater than about 3 kΩ to maintain a ratio greater than 50 between the two terms, and not be greater than about 8 kΩ to maintain a ratio greater than 10.

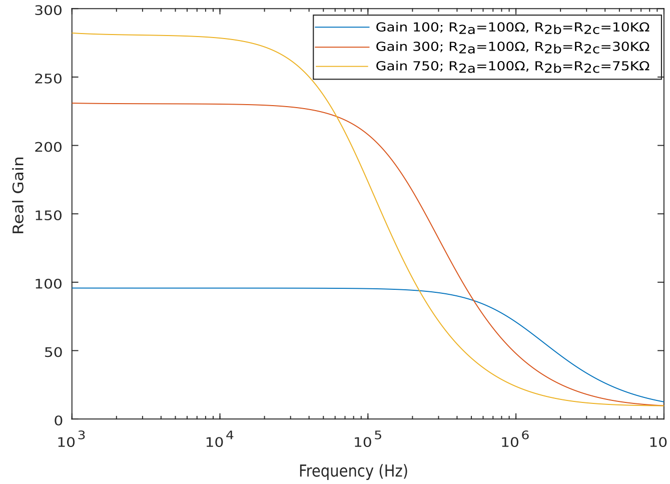

Fig. 20. Second-stage output according to different theoretical gain values requested.

Fig. 20 shows an example of the output of stage 2 indicating the gain of this second stage (circuit in Fig. 9), when different gains are requested. At this point, it is necessary taking into account (28).

As can be seen, choosing a resistor $R_{2a} = 100\ \Omega$, with $R_{2b} = R_{2c} = 75\ \mathrm{k}\Omega$, the theoretical (750) is reduced to around 280. The low R_{id} of the OA causes the gain to be attenuated. Two other cases are shown with values for R_{2b} of 30 and 10 kΩ. As can be seen, as the resistance R_b decreases, the gain approaches the theoretical gain (95% in the last case).

As already mentioned, for the design of the proposed second stage, it is important to know whether the filter to be implemented will be first order or second order. Remember that all the low-frequency noise (dc, sunlight, etc.) must be filtered out, and among them the 50–120-Hz noise from the lighting that can be received with enough signal strength to saturate, this stage or the next one should be considered.

We will first analyze theoretically and simulate a stage with first-order filtering (circuit in Fig. 9). For this amplification stage, a gain of 50 (V/V) is used. Fig. 21 shows the FT of the first two stages (first and second) with different HPF cutoff frequency values at second stage, achieved with capacitor C_{2a} values of 0.5, 3.5, and 6.8 μF.

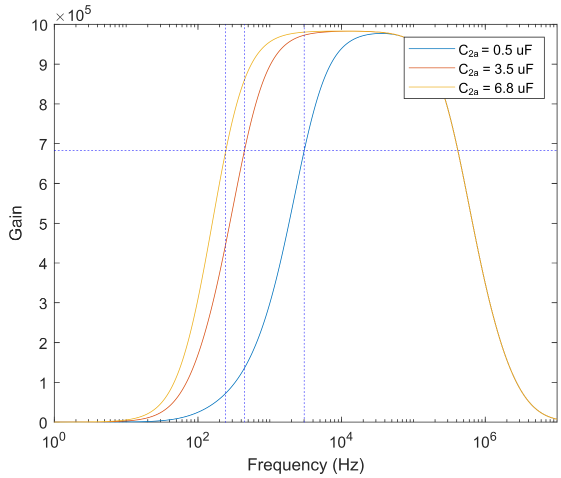

Fig. 21. Response of the second stage with first-order HPF at different capacitor values.

The results show that the corresponding cutoff frequencies are approximately 3 kHz, 320 Hz, and 210 Hz. It can be seen that for the lower cutoff frequencies, the ≈100-Hz noise is not sufficiently attenuated. The cutoff frequency must therefore be increased. In this case, the pass bandwidth with flat gain behavior is greatly reduced, not being sufficient for the application (in this case, frequencies between 5 and 14 kHz are proposed).

To also analyze the effect of v_{os} on the output if C_{2a} capacitor were not present, Fig. 22 illustrates the temporary result of amplifying in three different scenarios. The input signal from the PSD channel is 50-nA amplitude and the frequency 5 kHz (gain of the first two stages is 1M). The 100–120-Hz noise has not been included here to show only the effect of offset noise output at different gains.

As can be seen in Fig. 22, when the signal is amplified by 50 in the second stage and the capacitor C_{2a} is present, no information is lost, and the signal is centered at a value close to 0 V. However, in the absence of this capacitor, with an amplification of only 5 the offset grows enormously to -1.5 V (this is the result of the amplified input offset and I_{Bs} effects). The same happens with a gain of 10 which goes to values of -3 V. With amplifications higher than 15, the information would have been totally lost.

In selecting the components to implement the circuit in Fig. 9, the following values have been considered: $C_{2a} = 6.8\ \mu$ F, $R_{2a} = 100\ \Omega$, $R_{2b} = 5.05$ kΩ, and $R_{2c} = 5.05$ kΩ (selected from standardized values of commercial components).

In case a more pronounced attenuation of the low frequencies is required for this second stage, a second-order filter must be used. The authors put forth a proposal for a filtering stage based on the Sallen–Key topology (Fig. 7) whose TF is shown in (14). In this type of stage, it is challenging to separate the calculations of gain and cutoff frequency. Usually, some conditions are imposed and values are given to some components, with the remaining components being calculated. In this instance, it is proposed that capacitors of equal value be used, with the value of these capacitors then being assigned and the resistor values subsequently calculated.

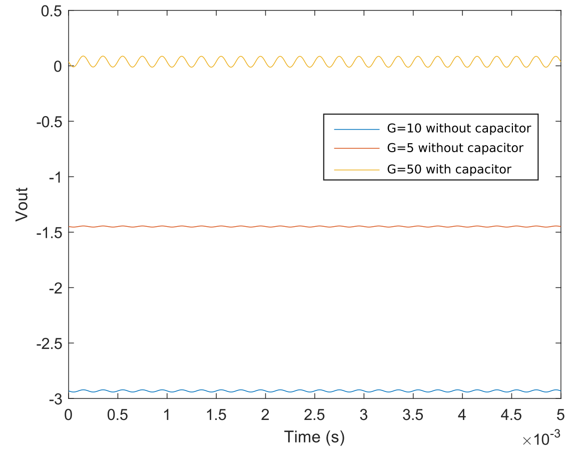

Fig. 22. Output signal in the second stage in the time domain with different gains.

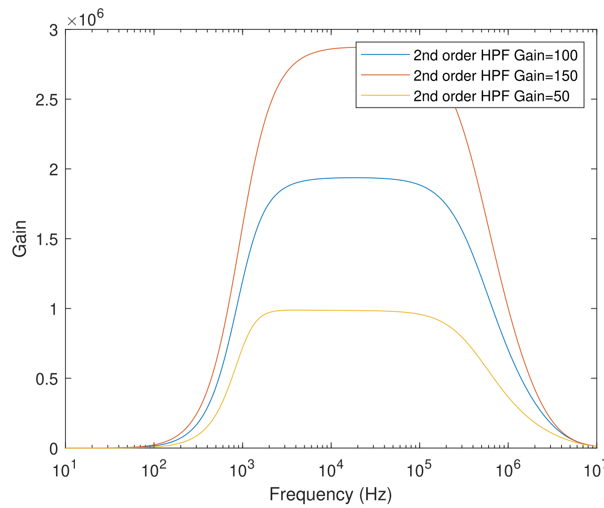

Fig. 23. Response of the first two stages when a second-order HPF is used in the second stage, for different gain values. The analysis was performed using a cutoff frequency of 950 Hz and 300-nF capacitors.

It should be noted that this type of circuit does not support a gain as high as the first order, because they reduce the passband. Fig. 23 demonstrates the response of the first two stages. The cutoff frequency has been set at 950 Hz and capacitors of 300 nF have been applied at different gain values (50, 100, and 150). As illustrated in Fig. 23 the high attenuation at 100-Hz frequency is readily apparent in a second-order Sallen–Key topology.

In this instance, when using this specific type of filter, it is advisable to operate at kHz frequencies with gains that are close to, but not exceeding, 50.

The application with a first- or second-order filter circuit will depend on the 50–120-Hz noise generated by lighting in the working environment. To continue the study, both in the simulated and empirical tests, we will use the second-order stage with a cutoff frequency of 750 Hz and a gain of 50. The proposed component values for the application are as follows: $C_{21} = C_{22} = 300$ nF, $R_{21} = 6.98$ kΩ, $R_{22} = 252\ \Omega$, $R_{23} = 221\ \Omega$, and $R_{24} = 10.8$ kΩ.

3) Third Stage: The third stage, as previously described, involves the final gain adjustment and bias addition to the signal prior to delivery to the ADC. In this instance, with

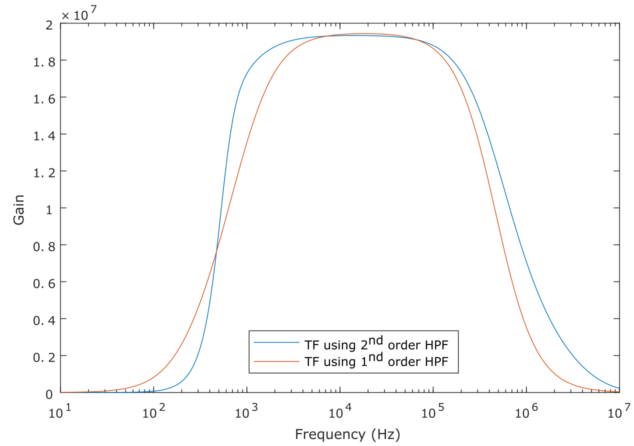

Fig. 24. Whole circuit response using a first-order HPF and second-order HPF in the second stage.

TABLE II
COMPONENTS VALUES USED WITH FIRST-ORDER HPF

$\mathbf{R_{11}}$	$\mathbf{C_{11}}$	$\mathbf{R_{12}}$	$\mathbf{C_{12}}$	$\mathbf{R_{2a}}$	$\mathbf{R_{2b}}$
298 Ω	6.8 μ F	20 kΩ	19 pF	100Ω	5.05 kΩ
$\mathbf{R_{2c}}$	$\mathbf{C_{2a}}$	$\mathbf{R_{31}}$	$\mathbf{R_{32}}$	$\mathbf{R_{33}}$	$\mathbf{C_{31}}$
5.05 kΩ	6.8 uF	2.05 kΩ	100 Ω	2.05 kΩ	6.8 μ F

a first amplification stage of 20l and a second amplification stage of 50, the required gain is 20. The proposed circuit is illustrated in Fig. 11.

When designing this third stage, two further aspects must be taken into account. The first is that some filtering element will also be included, mainly to remove the output offset noise from the second stage. The second is the need to include a bias voltage to obtain the maximum dynamic range in the input signal to the ADC. In our case, as signals between 0 and 3 V are desired at this input, a bias voltage of 1.5 V will be added.

Note that although different combinations of resistors can be used to achieve the same gain, it is advisable that they are not too large, to reduce the effect of offset noise and not too small, so that the current required is not too high. A compromise has to be reached, whereby the resistors are selected to prevent the current from exceeding 15 mA, if this is possible.

In both the cases, having incorporated either a first-order or a second-order filter stage, the component values for developing this third stage are shown in Table II. For its choice, it has been considered that a cutoff frequency of the HPF of 250 Hz is desired, adding an offset of 1.5 V and a gain of 20 (R_{32} = 100 Ω, R_{31} = 2.05 kΩ, C_{31} = 6.8 μ F, and $R_{33} = R_{31}$). In all the cases, these are components with standardized and commercial values.

Fig. 24 illustrates a comparison of the TFs of the entire system, for both the scenarios: using an amplifier circuit with either a first-order or second-order filter in stage 2.

To analyze the temporal behavior of the entire circuit, four signals of frequencies 5, 7, 9, and 11 kHz were used, in addition to a 100-Hz signal emulating lighting noise signal are introduced at the input of the first stage. Note that the 100-Hz signal is two orders of magnitude higher than the other signals which presents a very conservative situation. The output of the third stage is analyzed for the cases where first- and

TABLE III
COMPONENT VALUES USED WITH SECOND-ORDER HPF

$\mathbf{R_{11}}$	$\mathbf{C_{11}}$	$\mathbf{R_{12}}$	$\mathbf{C_{12}}$	$\mathbf{R_{21}}$	$\mathbf{R_{22}}$	$\mathbf{R_{23}}$
298 Ω	6.8 μ F	20 kΩ	19 pF	6.98 kΩ	252Ω	221Ω
$\mathbf{R_{24}}$	$\mathbf{C_{21}}$	$\mathbf{C_{22}}$	$\mathbf{R_{31}}$	$\mathbf{R_{32}}$	$\mathbf{R_{33}}$	$\mathbf{C_{31}}$
10.8 kΩ	300 nF	300 nF	2.05 kΩ	100 Ω	2.05 kΩ	6.8 μF

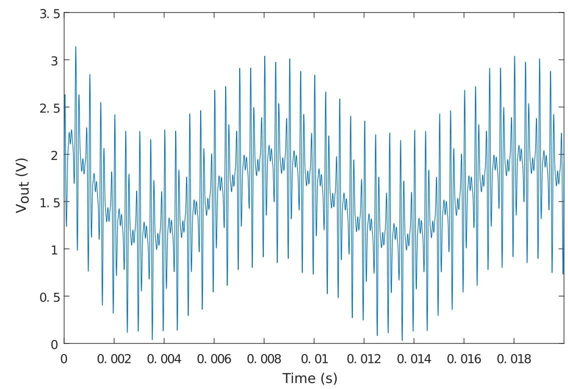

Fig. 25. Output signal in the third stage using first-order HPF in the second stage.

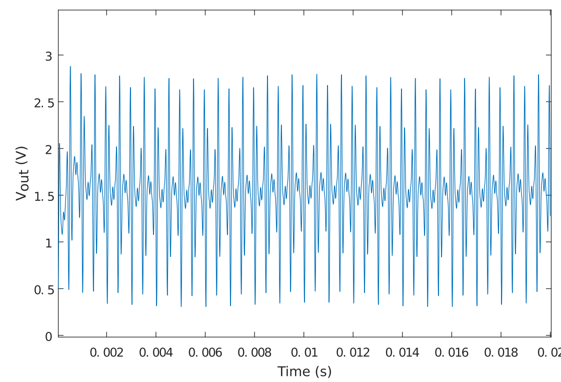

Fig. 26. Output signal in the third stage using second-order HPF in the second stage.

second-order filtering has been introduced in the second stage. In Fig. 25, it can be seen how a first-order filtering would not be sufficient for the (most likely) cases where there is 100-Hz noise coming from the luminaires. Then, even if filtered, given the total gain of the system (necessary for low PSD currents), the output may saturate and lose information.

Fig. 26 shows the same case, but now using a second-order filtering in the second stage. It can be seen how the remaining 100-Hz noise is almost negligible, thus being able to use the highest possible gain margin. It is evident that the first case has a great risk of reaching saturation.

V. EMPIRICAL RESULTS

This section shows the output signals of the proposed circuits, which have been mounted on a PCB, following the proposed methodology. It is important to note that the passive components of the design must have low tolerance values to achieve the same or as similar as possible gain in the four channels.

High-amplifier circuits using operational amplifiers are very sensitive to power supply noise. Therefore, it is necessary

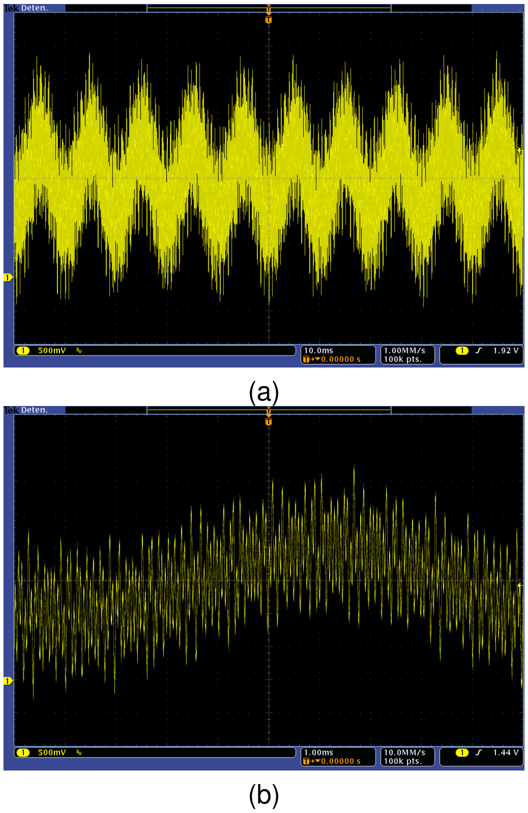

Fig. 27. Output signal with 100-Hz noise signal with a first-order high-pass filter using a sampling rate of (a) 1 MHz and (b) 10 MHz.

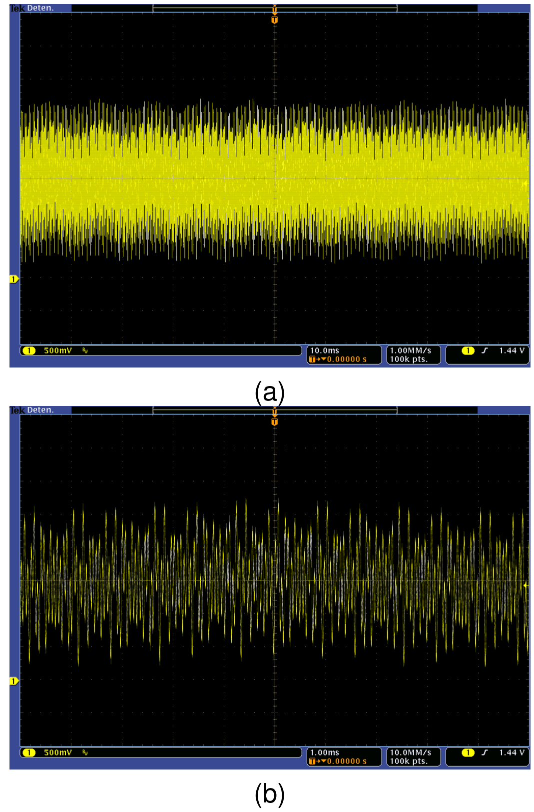

Fig. 28. Output signal with 100-Hz noise signal with a second-order high-pass filter using a sampling rate of (a) 1 MHz and (b) 10 MHz.

to supply the circuit with constant and filtered voltages. In addition, the digital and analog power supplies will be independent.

The values of the proposed components are shown in Tables II and III.

The test to verify the design is to capture the output signal from the conditioning system in a real working environment and analyze the SPAN range that the signal is covering. A real working environment could be composed of four LED emitters in a square arrangement (1 m between LEDs) as shown in Fig. 12. The receiver is placed below one of the emitters. The distance between the LEDs and the PSD receiver is 2.5 m to replicate the conditions of the simulations.

Each emitter transmits a sinusoidal signal of 5, 7, 9, and 11 kHz. Two configurations are analyzed using a first-order HPF and a second-order HPF.

In the second test, we check the performance of the design using a first-order HPF and a second-order HPF with a high 100-Hz noise signal reaching the receiver via reflections rather than directly. Fig. 27 shows the output signal using a first-order HPF. The 100-Hz noise signal is clearly visible in Fig. 27(a) and (b) (acquired using sampling rate of 1 and 10 Msamples/s, respectively). This causes the span to be higher than 3 V.

Fig. 28 shows the output signal using a second-order high-pass filter. Now, the 100-Hz noise signal is significantly reduced and the SPAN is below than 3 V. These results show that the circuit behaves analogous to the simulations. We have also seen the importance of properly filtering the 100-Hz signal being the best implementation the one using the second-order filter.

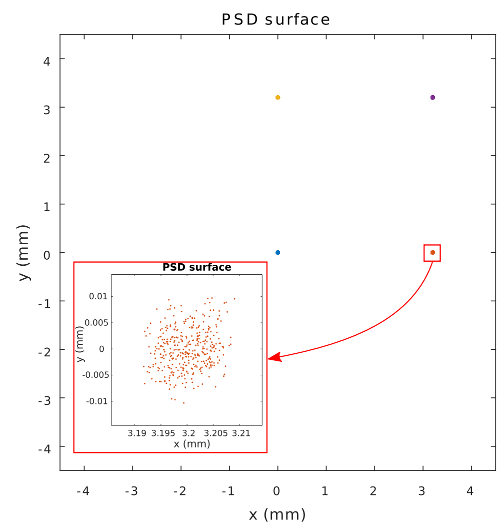

Fig. 29. Emitters' impact point on the PSD surface.

Fig. 29 is an example of a real positioning application using the amplification and conditioning circuit proposed. It shows the impact point of the four emitters (each one in a different color) on the PSD surface calculated with (1) and (2). Each emitter is correctly positioned.

Each signal of each channel of the PSD is processed to obtain the signal strength of each emitter. A Goertzel algorithm is used. Note that this part of the document is presented as a summary as a result of the final application, but is not part of the scope of this article. But, as can be seen in the enlargement of Fig. 29, the measurement accuracy is very high, which indicates that the conditioning circuit designed is of high quality. The dispersion would mean an error of 4 mm in the measurement at 2.5 m, despite the high amplification required, the very low input signal, and the high noise.

VI. Summary and Conclusion

This work has developed a procedure and technique for the development of signal conditioning circuits that work with optical sensors. Given the small signal delivered by these sensors, this work establishes the importance and need to amplify up to several million, reduce the noise bandwidth, eliminate any noise coming from sunlight or low-frequency lighting, add the optimum bias to the signals at the input of the ADCs taking full advantage of the span, and quantify the noise, among others.

The first step in the design process has been to know the specific characteristics of the devices to be used and the result to be achieved. For example, in the case of the optical sensor, it is necessary to know its wavelength response and sensitivity to calculate the signal it will deliver and its noise characteristics. It is also necessary to know the type of input of the ADC and its range to design the conditioning circuit with the best possible resolution and without risk of loss of information. From the working environment, it is necessary to know the model of the channel through which the signal will be transmitted to determine the signal power that will reach the optical sensor from the emitters.

Once the above assumptions have been established, the design of the conditioning circuits can begin. The first step is to determine the required gain and bandwidth of the circuit, as well as the types of noise and errors that will be present. It is also important to know that four exactly equal amplifier channels must be implemented, which implies a high quality of components, preferably chips with four OAs, and possibly a system calibration phase. In any case, the required gain will be very high, which will force the use of several stages, and great care will have to be taken with dc noise signals (output offset) and low-frequency signals from lighting systems. Three stages have been proposed in this design methodology.

As a general case for this type of sensors, it has been necessary to establish a first transimpedance stage with very high gain and bandpass filtering (which limits the noise at the working frequency and avoids instabilities). Next, an amplification stage with second-order filtering was suggested to complete the filtering of the low-frequency noise signals from lighting. And finally, a third stage for gain adjustment and biasing if necessary, to center the signal in the input range of the ADC. In all the cases, great care must be taken in the selection of component values to limit/eliminate the output clipping of the various stages due to dc offset.

It is obvious that the ideal OA should have a very high GBWP, no voltage and current noise, zero bias current, and zero offset voltage, as well as encapsulating all the OAs needed for the same stage of all the branches of the system. Since it is very difficult to meet all the requirements and to be ideal in several aspects, the proposed procedure has included an analysis of how to study the effects of nonideal values and how to calculate their true behavior to meet the requirements.

Both in the simulation phase and in the empirical testing phase, the proposed design procedure has been found to achieve the objectives in a very adequate way, obtaining the high gains required while avoiding saturation and loss of information due to multiple influencing factors that have been documented.

CRediT Authorship Contribution Statement

Research conceptualization, José Luis Lázaro-Galilea, Álvaro De-La-Llana-Calvo, and Carlos Andrés Luna-Vázquez; data curation, José Luis Lázaro-Galilea, Álvaro De-La-Llana-Calvo, Ignacio Bravo-Muñoz, and Carlos Cruz-De-La-Torre; formal analysis, Álvaro De-La-Llana-Calvo, José Luis Lázaro-Galilea, and Carlos Andrés Luna-Vázquez; funding acquisition, Alfredo Gardel-Vicente, José Luis Lázaro-Galilea, Ignacio Bravo-Muñoz, and Álvaro De-La-Llana-Calvo; investigation, José Luis Lázaro-Galilea, Alfredo Gardel-Vicente, Álvaro De-La-Llana-Calvo, Carlos Andrés Luna-Vázquez, Ignacio Bravo-Muñoz, and Carlos Cruz-De-La-Torre; methodology, José Luis Lázaro-Galilea, Álvaro De-La-Llana-Calvo, and Carlos Andrés Luna-Vázquez; project administration, Álvaro De-La-Llana-Calvo, José Luis Lázaro-Galilea, and Alfredo Gardel-Vicente; resources, José Luis Lázaro-Galilea, and Álvaro De-La-Llana-Calvo; supervision, and José Luis Lázaro-Galilea; validation, José Luis Lázaro-Galilea, Álvaro De-La-Llana-Calvo, Alfredo Gardel-Vicente, Carlos Andrés Luna-Vázquez, Ignacio Bravo-Muñoz, and Carlos Cruz-De-La-Torre; visualization, Álvaro De-La-Llana-Calvo, José Luis Lázaro-Galilea, and Carlos Cruz-De-La-Torre; writing—original draft and writing—review and editing, José Luis Lázaro-Galilea, Álvaro De-La-Llana-Calvo, Alfredo Gardel-Vicente, Carlos Andrés Luna-Vázquez, Ignacio Bravo-Muñoz, and Carlos Cruz-De-La-Torre. All authors read and agreed to the published version of the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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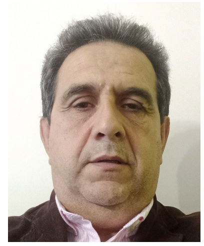

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